

"Chapter- 9"

"Morphological

Image processing"

Mathematical morphology is a tool for extracting image components that are useful in the representation and description of the image.

Morphology applies certain simple rules & shapes to process images. Its objective is to identify features of the image.

NOTE

set space of binary image is 2^2 . Each element of a set is a 2D vector whose coordinates are (x, y) of a black/white (depending on convention) pixel.

set space of grey level image is 2^3 .

Each element of the set is a 3D vector (x, y) and intensity level.

Set theory

$a \in A$: $a = (a_1, a_2)$ is an element of A

$a \notin A$: not an element of A

$A = \{(a_1, a_2), (a_3, a_4)\}$ set representation

$A = \emptyset$ empty set

$A \subseteq B$ subset $A \subset B$ ^{proper} subset $[A \subset B]$
 $[A \subseteq B]$

$$C = A \cup B \quad \text{union}$$

$$D = A \cap B \quad \text{intersect}$$

$$A \cap B = \emptyset \quad [\text{disjoint / mutually exclusive}]$$

$$B = \{w \mid w = -b, \text{ for } b \in B\} \quad \text{Reflection}$$

$$A^c = \{w \mid w \notin A\} \quad \text{complement}$$

$$(A)_z = \{w \mid w = a + z, \text{ for } a \in A\} \quad \text{translation}$$

$$A - B = \{w \mid w \in A, w \notin B\}$$

$$= A \cap B^c \quad \text{difference}$$

Structuring element

It is a small image used as a moving window. Ex - diamond shape, cross shape, box shape, line shape.

In SE we can alter the central pixel unlike a filter/kernel.

Fit $\rightarrow$ All ON pixels in the SE covers ON pixels on the image.

Hit $\rightarrow$ Any ON pixel in the SE covers an ON pixel in the image.

Miss $\rightarrow$ No ON pixel in the SE covers an ON pixel in the image.

Erosion $\ominus$

Shrinks the object

foreground shrinks

Background enlarges

smoothes object boundaries & removes small objects.

$$A \ominus B = \{ z \mid (B)_z \subseteq A \}$$

Set of all points z , such that B translated by z is contained in A .

$$g(x, y) = \begin{cases} 1 & \text{if } s \text{ fits } f \\ 0 & \text{otherwise} \end{cases}$$

new pixel value is determined by $g(x, y)$.

Dilation $\oplus$

grows the object

Enlarges foreground

shrinks background

smoothes the object boundaries

$$A \oplus B = \{ z \mid (\hat{B})_z \cap A \neq \emptyset \}$$

all z in A such that B hits A when

origin of $B = z$ such that overlap A by

at least one element.

new pixel value is determined by :

$$g(x,y) = \begin{cases} 1 & \text{if } s \text{ hits } f \\ 0 & \text{otherwise} \end{cases}$$

Edge detection

Sub input image from dilated image, so that edge remains.

Sub eroded image from the input image so that edge remains.

NOTE

Dilation & erosion are duals of each other.

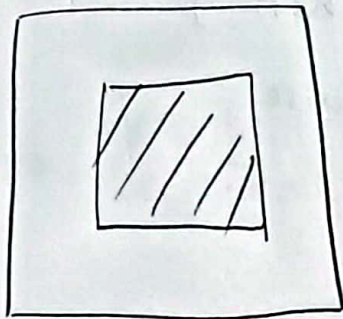
$$(A \ominus B)^c = A^c \oplus \hat{B}$$

complement reflection

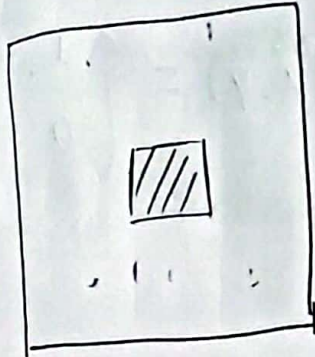
For symmetric SE :

$$(A \oplus B)^c = A^c \ominus \hat{B}$$

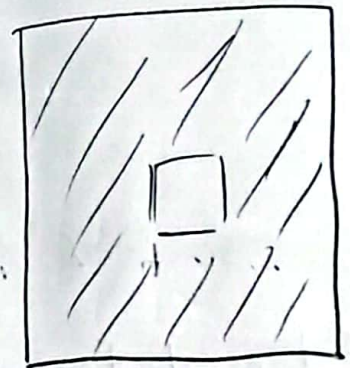
Duality



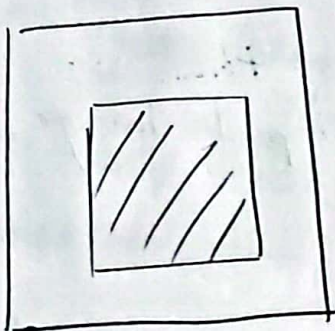
A



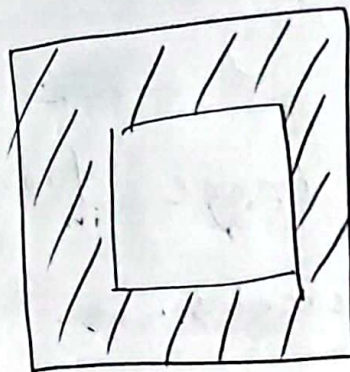
$A \ominus B$



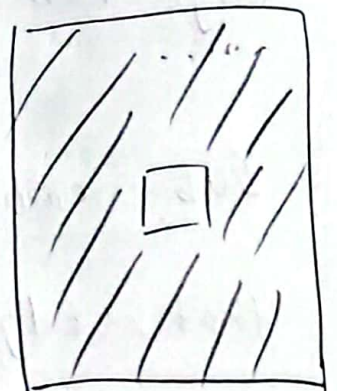
$(A \ominus B)'$



A



A'



$A' \oplus \hat{B}$

Opening

Erosion followed by dilation.

$$A \circ B = (A \ominus B) \oplus B$$

Note

Smoothed the outline by rounding off any sharp points.

Removes any part that is smaller than SE.

Disconnect / open any thin bridges.

Does not remove holes or gaps.

Does not make the basic core size of the shape larger / smaller.

$$\text{Boundary Extraction} = A - (A \ominus B)$$

Closing

Dilation followed by an erosion.

$$A \bullet B = (A \oplus B) \ominus B$$

Smoothed the outline by filling in any holes and indentations. It will also form connecting bridges to other shapes that are close enough for SE. Does not make the basic core size of the shape larger/smaller.

NOTE

opening is dual of closing.

Opening foreground pixels with particular SE

= Closing the background pixels with the same element.

Hit or miss transformation

SE can contain a pattern of 3 types

1 $\rightarrow$ foreground

0 $\rightarrow$ background

x $\rightarrow$ don't care

Ex-1

shape $\rightarrow$

1	1	1
x	1	x
x	0	x

NOTE

If the foreground and background pixels in the structuring element exactly match foreground and background pixels in the image then the pixel underneath the origin of the structuring element is set to the foreground color.

SE to represent corners

x	1	x
0	1	1
0	0	x

x	1	x
1	1	0
x	0	0

x	0	0
1	1	0
x	1	x



0	0	x
0	1	1
x	1	x



$$A \circledast B = (A \ominus B_1) - (A \oplus \hat{B}_2)$$

NOTE

background is not necessary to detect disjoint sets.
Only simple erosion is enough.

* Thinning and thickening

Thinning is used to remove selected foreground pixels from binary images. It is used for thinning lines to 1px width.

$$A \otimes B = A - (A \circledast B) = A \cap (A \circledast B)^c$$

$$\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$$

B^i is rotated version of B^{i-1}

NOTE

If foreground and background fit the SE exactly, then the pixel at the origin of the SE is set to 0.

SE at the origin is 1 / don't care.

Ex-1

0	0	0
1	1	1
0	1	0

B1

1	0	0
0	1	0
1	0	1

after clockwise rotation

B2

0	1	0
1	1	0
0	1	0

B3

B^n here n is the no. of neighbour

STON

Thickening

Used to grow selected regions of foreground pixels. Ex - Convex hull.

$$A \odot B = A \cup (A * B)$$

$$A \odot \{B\} = ((\dots ((A \odot B^1) \odot B^2) \dots) \odot B^n)$$

If foreground and background match exactly the SE, then set the pixel at its origin to 1!
SE at the origin is 0 or don't care.

$$A \odot B = A \cup (A * B)$$

A

$$X = A^c$$

$$Y = X \otimes B$$

$$Z = Y^c$$

Ignoring the disjoint point we can see that

$$Z = A \odot B$$

————— END of chap 9 —————